

# Advanced Pandas vs SQL Window Functions Cheat Sheet

| SQL Window Function                                       | Pandas Equivalent                                   |
|-----------------------------------------------------------|-----------------------------------------------------|
| ROW_NUMBER() OVER(PARTITION BY col ORDER BY val)          | df.sort_values('val').groupby('col').cumcount() + 1 |
| RANK() OVER(PARTITION BY col ORDER BY val)                | df.groupby('col')['val'].rank(method='min')         |
| DENSE_RANK() OVER(PARTITION BY col ORDER BY val)          | df.groupby('col')['val'].rank(method='dense')       |
| SUM(val) OVER(PARTITION BY col ORDER BY ...)              | df.groupby('col')['val'].cumsum()                   |
| AVG(val) OVER(PARTITION BY col)                           | df.groupby('col')['val'].transform('mean')          |
| LAG(val,1)                                                | df.groupby('col')['val'].shift(1)                   |
| LEAD(val,1)                                               | df.groupby('col')['val'].shift(-1)                  |
| FIRST_VALUE(val)                                          | df.groupby('col')['val'].transform('first')         |
| LAST_VALUE(val)                                           | df.groupby('col')['val'].transform('last')          |
| Moving Average (ROWS BETWEEN 2 PRECEDING AND CURRENT ROW) | df['val'].rolling(3).mean()                         |